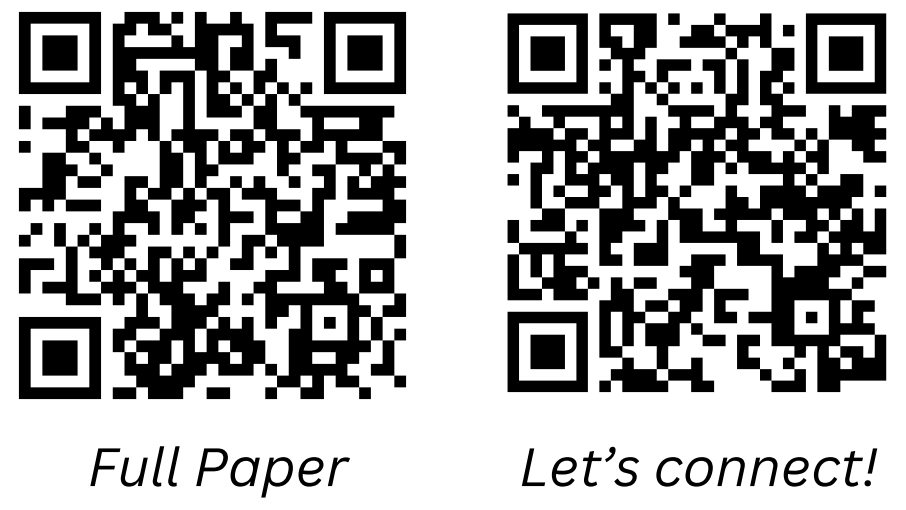


Intrinsic behavioral dispositions shape the dynamical complexity of opinion evolution in adaptive social networks



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Background: Opinion-dynamics research has traditionally focused on aggregate outcomes—such as consensus, polarization, fragmentation, or network clustering—often using static snapshots of the system. This can obscure how opinions evolve over time at the level of individual agents and whether different behavioral tendencies produce systematically different experiences of stability, volatility, and unpredictability.

Approach: We simulated an adaptive social network in which agents differ in three behavioral dispositions: **homophily**, **neophily**, and **social conformity**. For each node, we recorded its opinions over the course of the simulation and computed the normalized Lempel–Ziv complexity (nLZ) of the resulting trajectory to quantify its temporal unpredictability.

Results: Across homogeneous, dual-faction, and fully heterogeneous network settings, each agent type exhibited a persistent complexity signature: homophilic agents became increasingly unpredictable, neophilic agents maintained low and stable complexity, and conformic agents showed a two-phase pattern. By changing a focal node’s behavioral disposition while holding its network state and social environment fixed, we further showed that these profiles are causally driven by internal behavioral rules, rather than merely correlated with network composition.

Conclusion: Behavioral dispositions act as dynamical fingerprints that causally shape how individuals experience opinion stability and unpredictability.

The Model

$$\frac{dx_i}{dt} = c(\langle x \rangle_i - x_i) + \epsilon \quad (1)$$

$$\frac{dw_{ij}}{dt} = hF_h(x_i, x_j) + aF_a(\langle x \rangle_i, x_j) \quad (2)$$

$$\langle x \rangle_i = \frac{\sum_j w_{ij} x_j}{\sum_j w_{ij}} \quad (3)$$

$$F_h(x_i, x_j) = \theta_h - |x_i - x_j| \quad (4)$$

$$F_a(\langle x \rangle_i, x_j) = |\langle x \rangle_i - x_j| - \theta_a \quad (5)$$

where, a attention to novelty (neophily)
 c social conformity
 h homophily

$x_i \in \mathbb{R}$ opinion of the i^{th} agent

w_{ij} flow of information from source node j to target node i

$\langle x \rangle_i$ the social norm, i.e., the weighted average of opinions in node i ’s neighborhood

F_h designates the strengthening of edge connecting j to i when nodes i and j have similar opinions (i.e., homophily)

F_a designates the strengthening of the same edge when node j has an opinion different from that of the perceived social norm of node i (i.e., novelty)

Lempel-Ziv (LZ) complexity

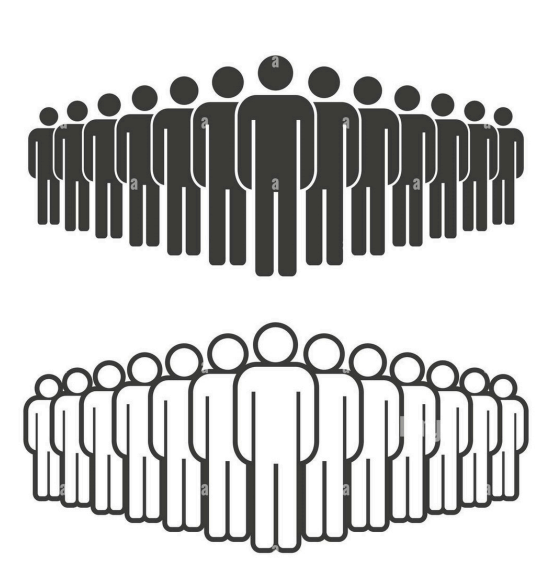
- LZ measures how repetitive or compressible a sequence is.
- LZ is computed by scanning a sequence and counting the number of new, previously unseen patterns it contains.
- E.g., $0 \cdot 1 \cdot 10 \cdot 010 \cdot 1101 \cdot 100100 \cdot 110 \rightarrow \text{LZ} = 7$

More repetition, lower complexity.
More novelty, higher complexity.

Experimental setup

Scenario 1

Pure



All agents share the same dominant behavioral disposition.

Scenario 2

50-50



Two equally sized groups with different dominant dispositions interact.

Scenario 3

Mixed

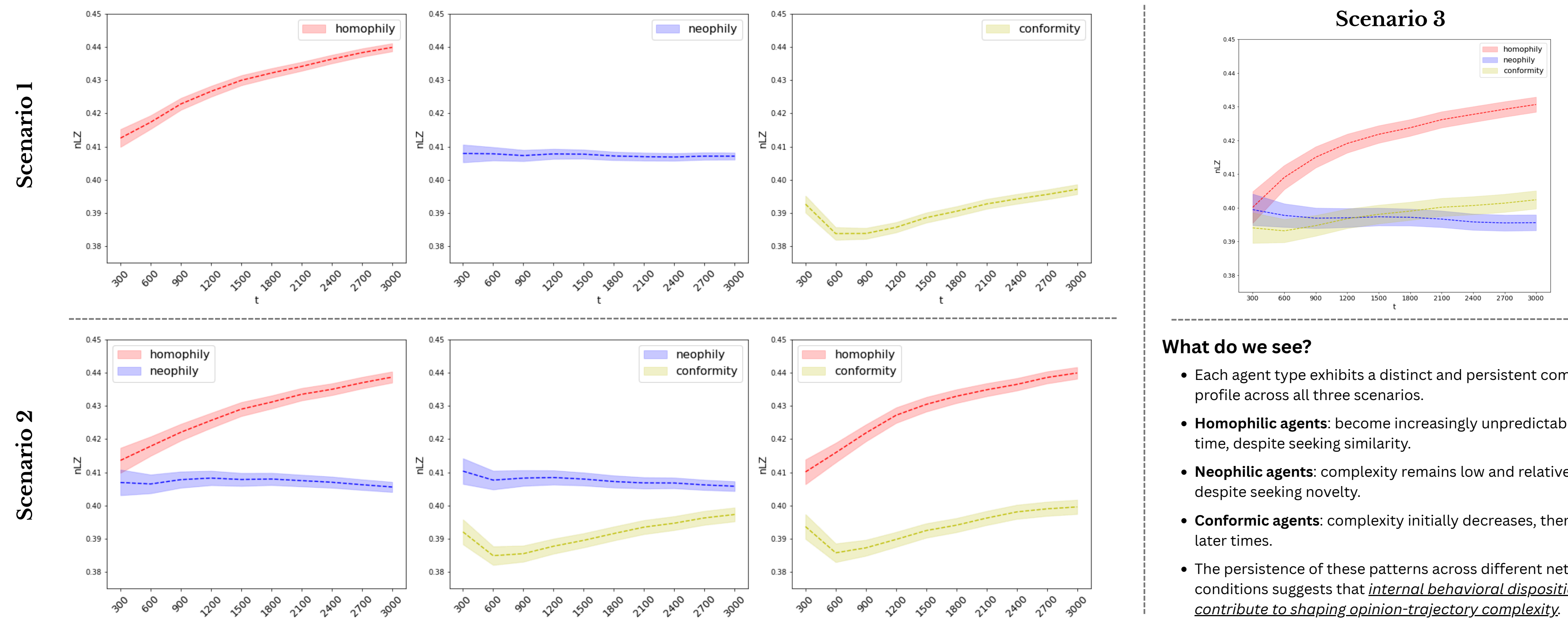


Agents have continuously varying behavioral parameters, creating a fully heterogeneous population.

How do we define agent types?

- Scenarios 1 & 2
- Homophilic: $a, c \sim N(0.05, 0.025)$; $h \sim N(0.25, 0.025)$
 - Neophilic: $h, c \sim N(0.05, 0.025)$; $a \sim N(0.25, 0.025)$
 - Conformic: $h, a \sim N(0.05, 0.025)$; $c \sim N(0.25, 0.025)$
- Scenario 3
- All agents: $a, c, h \sim U(0.05, 0.3)$

Results



What do we see?

- Each agent type exhibits a distinct and persistent complexity profile across all three scenarios.
- Homophilic agents:** become increasingly unpredictable over time, despite seeking similarity.
- Neophilic agents:** complexity remains low and relatively stable, despite seeking novelty.
- Conformic agents:** complexity initially decreases, then rises at later times.
- The persistence of these patterns across different network conditions suggests that internal behavioral dispositions contribute to shaping opinion-trajectory complexity.

Testing causality: isolating internal disposition from external context

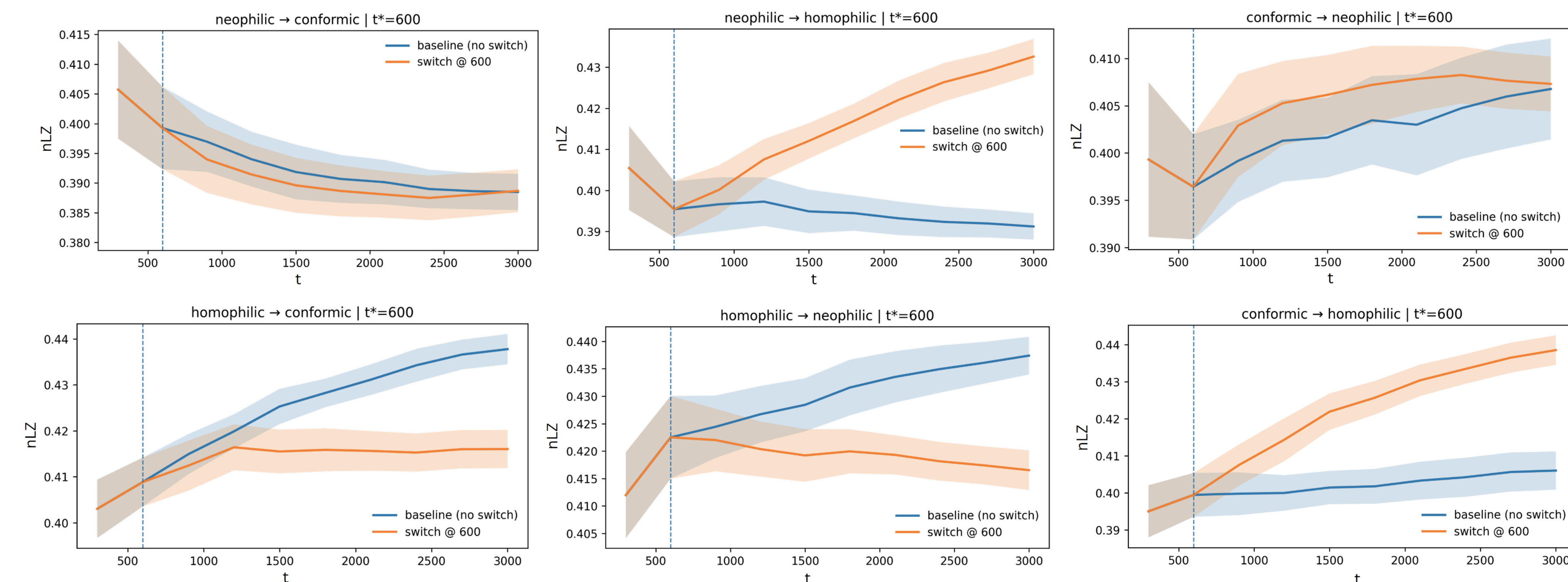
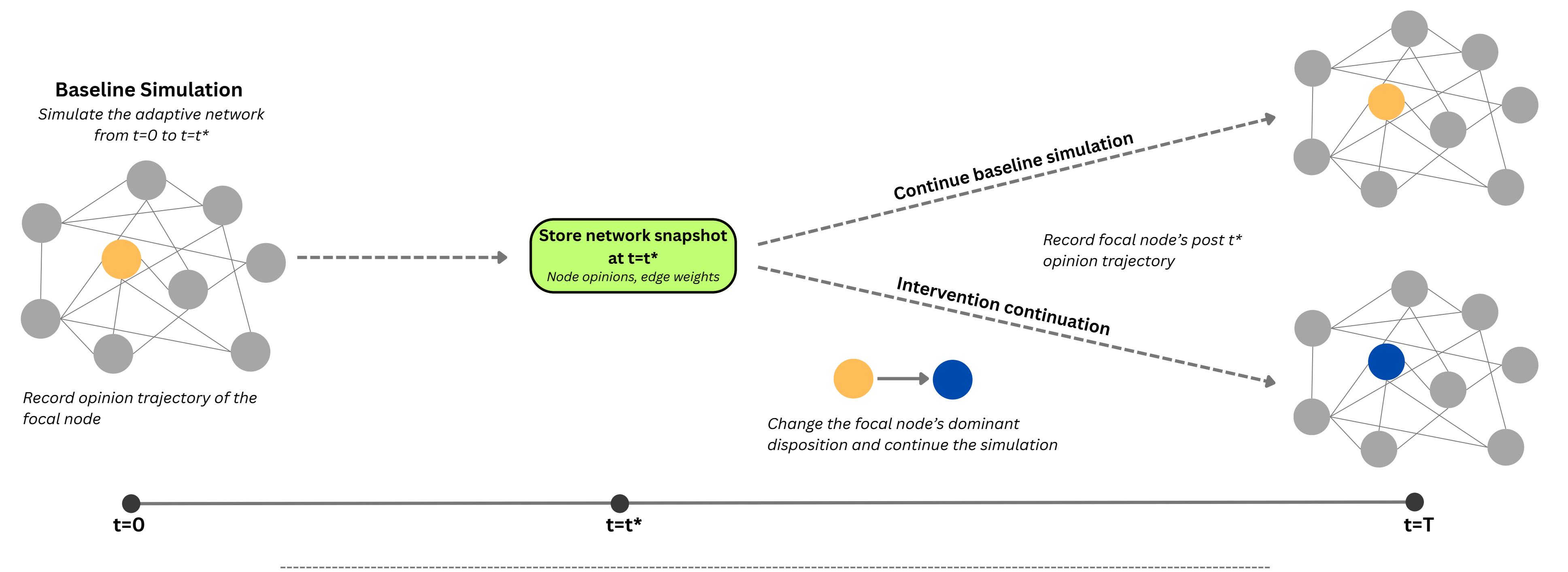


Figure: Causal intervention results for Scenario 3 (heterogeneous environment) at intervention time $t^* = 600$.

For each transition, the nLZ complexity of the focal node’s opinion trajectory is shown for the **baseline** (no intervention; in blue) and **intervention** (in orange) cases, averaged across nodes and runs with 95% confidence intervals.

Summary

- We compared pure, 50–50, and fully mixed networks to test whether behavioral signatures persist across different social environments.
- For each agent, we measured the nLZ complexity of its opinion trajectory to quantify how unpredictable its opinions became over time.
- Homophilic agents became increasingly unpredictable, neophilic agents remained relatively stable, and conformic agents showed an initial decrease followed by later growth in unpredictability.
- These counterintuitive signatures remained consistent across all three network settings.
- In causal intervention tests, changing only a focal node’s internal disposition produced systematic shifts in its nLZ trajectory, showing that the observed profiles are causally driven by behavioral rules, not merely correlated with network composition.
- Future work will identify the network mechanisms underlying these patterns.

References

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